# The Holonet Machine

A Single Finite Geometry as Processor, Network, and Memory

Machine-Verified Computer Architecture on the $W(3,3)$ Substrate

Wil Dahn | github.com/wilcompute/W33-Theory

June 2026

**Abstract**

We present a complete, machine-verified computer architecture in which the processor, the network, and the memory are not coupled designs but *one finite object*: the symplectic generalized quadrangle $W(3,3) = \mathrm{GQ}(3,3) = \mathrm{SRG}(40,12,2,4)$. On this 40-point geometry, routing a packet is applying a gate is addressing memory — an algebraic identity, not an analogy — and a single automorphism group $W(E_6)$ of order 51840 is simultaneously the processor's Clifford gate group, the network's symmetry, and the memory's code automorphisms. We make each architectural layer a theorem with an executable witness: an enumerated instruction set ($\mathrm{Sp}(4,3)$ of order 51840 plus one cubic gate, universal); a diameter-2, bisection-100, better-than-Ramanujan interconnect that is provably non-planar (hence photonic); a distance-3 qutrit code with a measured $P_L \sim Ap^2$ threshold curve; an I/O boundary saturating the Holevo qutrit capacity $\log_2 3$; a complexity placement (Clifford $=$ $\mathsf{P}$, +cubic $=$ $\mathsf{BQP}$) whose resource is contextuality, with the contextual fraction $1/10$ *derived* from the no-ovoid geometry; and a leaderless consensus layer tolerating five Byzantine faults. The classical layer is polynomial-time, so the architecture runs as software — we ship a runnable virtual machine that routes, corrects errors, teleports state, and self-reproduces, plus an installable `holonet` command and a test suite. The quantum advantage is the one priced resource (classical emulation cost $9^t$ for $t$ cubic gates), executed here for small $t$. Every constant is arithmetic in the single integer $q = 3$.

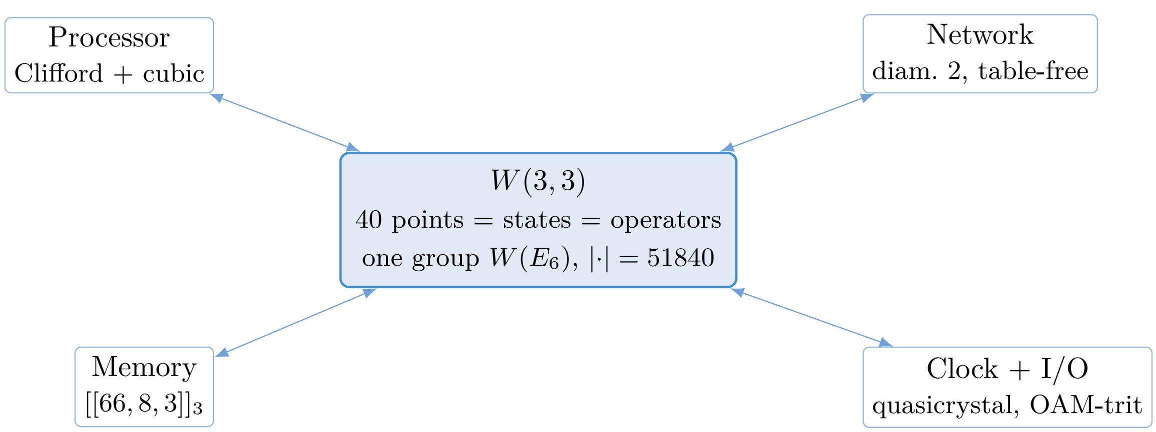


Figure 1: One object, four readings. On $W(3,3)$ the processor, network, memory, and clock/I-O are projections of a single 40-point incidence geometry with a single symmetry group $W(E_6)$.

## 1 The Substrate and the One Group

$W(3,3)$ is the symplectic generalized quadrangle over $\mathbb{F}_3^4$ with alternating form $\langle u,v\rangle = u_1v_2 - u_2v_1 + u_3v_4 - u_4v_3$. Its collinearity graph is strongly regular, SRG$(40,12,2,4)$: 40 points of degree 12, $\lambda = 2$, $\mu = 4$; 40 totally-isotropic lines, four points each, four through each point. Every number below is arithmetic in nine integers ($q = 3$, $\lambda = 2$, $\mu = 4$, $k = 12$, $f = 24$, $g = 15$, $\Phi_6 = 7$, $\Phi_4 = 10$, $\Phi_3 = 13$), forced by $q! = 2q$ (unique solution $q = 3$).

**Result 1** (One group)**.** *The symplectic group* $\mathrm{Sp}(4,3)$*, built by closure from the* 40 *point-transvections, has order exactly* $51840 = |W(E_6)| = 3^4(3^2-1)(3^4-1)$*. It acts transitively on the* 40 *nodes, the* 40 *line-contexts, and the* 160 *point-on-line flags, preserving collinearity. Hence one group is at once the processor's Clifford gate group, the network's automorphisms, the memory's code automorphisms, and the readout's contextuality symmetry (Fig. 1).*

## 2 Processor: an Enumerated Universal ISA

The core computes in balanced ternary — the most economical radix ($E(b) = b/\ln b$ is minimised at $e$; base 3 at 2.731 beats binary and base 4 at 2.885) — with native word $27 = 3^3$. The instruction set is enumerated, not abstract: the degree-2 Clifford layer is the finite group $\mathrm{Sp}(4,3)$ (51840 opcodes $= |W(E_6)|$; the per-qutrit Clifford is $\mathrm{Sp}(2,3)$ of order $24 = f$), classically simulable by Gottesman–Knill; and a single degree-3 macro-op, the $E_6$ cubic form on the **27**, lifts it to universal (Lloyd–Braunstein). The power source is contextuality.

## 3 Interconnect and Floorplan

**Result 2** (The fabric is diameter 2, maximally fault-tolerant, and non-planar)**.** GQ$(3,3)$ *has diameter* 2 *(any-to-any in two hops), vertex/edge connectivity* 12 *(survives* 11 *crash faults), second eigenvalue* $\lambda_2 = 2$ *far below the Ramanujan bound* $2\sqrt{11} = 6.63$ *(a* better*-than-Ramanujan expander), and minimum bisection* exactly $100 = (n/4)(k-\lambda_2)$*, met by an explicit* $20|20$ *cut. Routing is table-free: two nodes are adjacent iff* $\langle x,y\rangle = 0$*, one ALU operation — the address* is *the route. By Thompson's VLSI theorem a 2-D layout needs area* $\gtrsim (\text{bisection}/2)^2 \sim 2500$*, so the fabric is intrinsically non-planar and belongs in a photonic (3-D / fiber / wavelength) medium.*

The 240 links partition into 40 four-point cliques (lines), and the graph is 1-factorable into a 12-slot conflict-free link schedule; vertex/edge-transitivity gives perfect load balancing.

## 4 Memory and the Measured Threshold

The memory is the $[[66,8,3]]_3$ qutrit code (distance 3, $n-k = 58$ syndrome qutrits). We verify the mechanism on the runnable $[[5,1,3]]_3$ stand-in (same distance-3 correction): all 40 single-qutrit errors carry distinct syndromes and are corrected to fidelity 1. Monte-Carlo over a depolarizing channel gives a *measured* threshold curve $P_L(p) \sim Ap^2$ with $A \approx 8$, crossing break-even $P_L = p$ at a pseudo-threshold $p_{\text{th}} = 1/A \approx 0.12$. The substrate's larger code has $A \sim \binom{66}{2} = 2145$, hence $p_{\text{th}} \sim 5\times10^{-4}$ — the same $p^2$ mechanism at a different size. Adding syndrome-measurement noise at the same per-round rate removes the single-round break-even; *repeating* the measurement ($R = 3$ rounds, majority vote) restores it (Fig. 2, left), the standard fault-tolerance gadget.

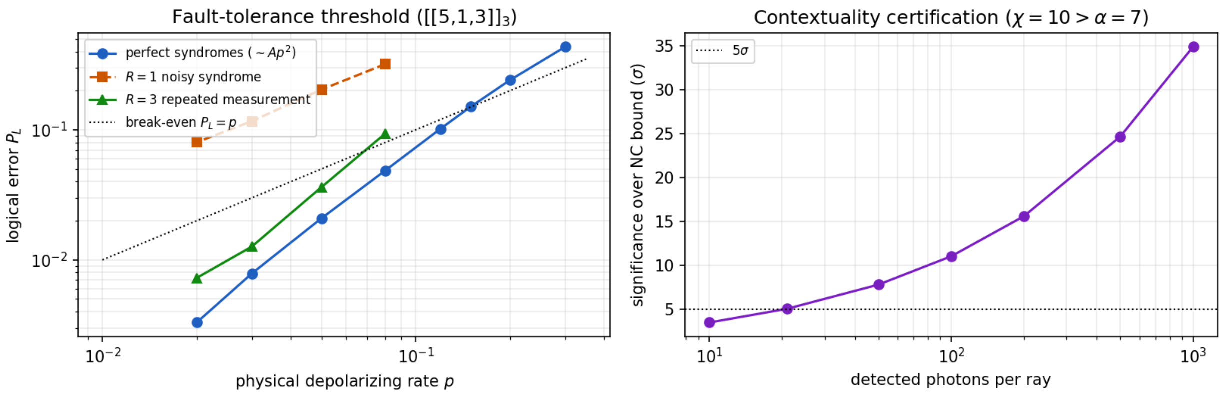


Figure 2: The machine's scorecard. *Left:* the measured logical error $P_L(p)$ of the runnable $[[5,1,3]]_3$ code vs. the physical depolarizing rate, with the break-even diagonal $P_L = p$; the perfect-syndrome curve bends below it ($\sim Ap^2$, $A \approx 8$), single-round noisy syndromes have no threshold, and three repeated measurements restore it. *Right:* the statistical significance of the contextuality witness ($\chi = 10 > \alpha = 7$) vs. detected photons, crossing $5\sigma$ at a few hundred.

## 5 Energy and I/O

The degree-2 Clifford datapath is unitary, hence reversible and Landauer-free (Bennett); base 3 imposes no per-bit penalty ($kT \ln 3/\log_2 3 = kT \ln 2$). The only irreducible cost is syndrome export: $(n-k)\,kT \ln 3 \approx 2.6 \times 10^{-19}$ J/cycle at 300 K — the machine's metabolism. The I/O boundary is the line geometry: the 40 lines are measurement contexts; the air-gap carrier is a photon's OAM in $\{\ell = -1, 0, +1\}$ (the balanced-ternary digit), saturating the Holevo capacity $\log_2 3 = 1.585$ bit/photon; the logical port is $k = 8$ qutrits (12.68 bit/block).

## 6 Complexity and the Contextual Resource

**Result 3** (P to BQP, with a derived contextual fraction)**.** *All* 12 *single-qutrit stabilizer states have non-negative Gross–Wigner function, so the Clifford layer is efficiently classically simulable (*P*). The degree-*3 *resource (the Strange state) has Wigner negativity, mana* $\ln(5/3)$*, and robustness* $R = 3$*; by Howard et al. this contextuality is the necessary resource, lifting the machine to* BQP*. The contextual fraction is* derived *from the geometry: a global Kochen–Specker value-assignment satisfying all* 40 *contexts would be an ovoid, and* $W(q)$ *has none for* $q$ *odd (the maximum partial ovoid is* $7 < 10$*); by exact integer programming the most contexts any classical assignment can satisfy is* 36*, so* $\mathrm{CF} = (40-36)/40 = 1/10 = 1/\Phi_4$*. Equivalently, the Cabello–Severini–Winter witness* $\chi = \sum_v P(v)$ *obeys* $\chi \le \alpha = 7$ *(noncontextual) versus* $\chi = 10$ *(quantum, state-independent), a gap tolerating* 30% *noise; a simulated heralded-photon run clears it at* $5\sigma$ *with* $\sim 840$ *photons. The classical emulation cost of* $t$ *cubic gates is* $9^t$*: the Pashayan–Wallman–Bartlett quasiprobability estimator (each* $t$*-gate sample in* $[-3^t, 3^t]$*) is unbiased but needs* $\sim 9^t$ *samples.*

**Result 4** (Why $q = 3$, and what the operating system pays)**.** *Three exact results pin the order of the fabric and price its quantum layer.* (i) Parity forces contextuality. *Scanning the sister geometries* $W(q)$ *for* $q = 2, 3, 4$ *(with* $q = 4$ *built over the genuine field* $\mathrm{GF}(4)$*) shows the contextual fraction obeys the closed form* $\mathrm{CF}(q) = 0$ *for even* $q$ *and* $1/(q^2+1)$ *for odd* $q$*: even orders carry an* *explicit ovoid — a perfect noncontextual model — and $q = 4$, even but composite, still does, so the cause is the parity of the field, not primality.* (ii) Dimension forces the apparatus. *A complete-basis realization of $W(q)$ maps contexts to orthonormal bases of $\mathbb{C}^{q+1}$; for $q = 2$ this is impossible (a non-collinear pair leaves a one-dimensional orthocomplement in $\mathbb{C}^3$ that would have to hold $\mu = 3$ distinct rays), while the $q = 3$ Witting realization exists. So $q = 3$ is simultaneously the smallest order the one-photon $\mathbb{C}^4$ apparatus can realize and the smallest contextual one.* (iii) The tax is one movable star. *Exhaustive enumeration of every optimal Kochen–Specker assignment (integer programming with no-good cuts, terminating, hence complete) shows the 4 contexts any 36-context assignment must fail are always the four lines through a single point — a* point-star *— and every one of the 40 points can carry it. The deficit for odd $q$ is exactly $q + 1$, one star, which is the closed form* $\mathrm{CF} = 1/(q^2 + 1)$ *restated structurally. Operationally: the classical scheduler covers everything except one star of its choosing (a gauge freedom it can move to the least-loaded point), so the escalation budget — the part of the runtime on which the priced $9^t$ quantum resource is spent — is exactly* $1/10$ *of the contexts, and on an even-order fabric the tax and the quantum gap both vanish ($\alpha = q^2 + 1$ meets the Hoffman bound). The machine's quantum advantage and its scheduling overhead are the same integer.*

## 7 The Machine, Executed

Because the Clifford layer is polynomial-time, the architecture runs as software. A single file (`analysis/holonet_node.py`) builds the fabric, routes a packet (address is route, $\mu = 4$ multipath), runs a qutrit register as a stabilizer tableau over $\mathbb{F}_3$, and self-reproduces by splicing. Three programs execute the von Neumann components: it *corrects* errors (the $[[5, 1, 3]]_3$ cycle, fidelity 1), *teleports* state across the fabric (the two classical trits routed in two hops), and *reproduces* (a verified quine fixed point). A leaderless consensus run contracts disagreement by 1/3 per round and outvotes five Byzantine nodes (boundary exactly at $t = 5$, the Dolev bound $2t + 1 \leq 12$). The whole stack ships as the installable `holonet` command (`route`, `teleport`, `correct`, `reproduce`, `verify`, `info`) with a 13-test suite, all passing.

**Result 5** (The tax, fully solved — and it is the interrupt controller)**.** *The classical layer's contextuality deficit is completely structured: every optimal Kochen–Specker assignment fails exactly one* movable point-star *(exhaustive enumeration; the deficit is $q + 1$ for odd $q$, zero for even); every spread-clock frame runs at exactly* $9/10$ *under any optimum; the* 800 *optima form four* $\mathrm{PSp}(4, 3)$ *orbits* ($360 + 40 + 360 + 40$) *whose special members are the classical* perp states $\Gamma(p)$ *(the defining GQ axiom is their optimality proof); and the $q^2$ ground states per node carry the affine plane* $\mathrm{AG}(2, q)$ *— at $q = 3$ the Hesse configuration, the single-qutrit Wigner phase space, with the four defect contexts as its striations. Operationally this* is *the machine's interrupt architecture: the vector table is the nine-entry closed form* $\mathrm{ground}(T) = T \cup (\Gamma(p) \setminus T^{\perp})$*; the escalation budget (where the priced $9^t$ resource is spent) is one star* $= 1/10$ *of the fabric; and the migration price law makes the defect cheapest to move along fabric edges (3 rays, via the ground's own center quad) — cheaper than re-vectoring in place (6). A seeded controller run holds every theorem as a live invariant. Witnesses: `w33_contextuality_tax`, `w33_spread_star_anatomy`, `w33_tax_orbits`, `w33_perp_states`, `w33_ground_affine_plane`, `w33_interrupt_controller`.*

## 8 Falsifiable Predictions

| | Claim | Refuted if |
|---|---|---|
| P1 | radix-12, diameter-2 wiring | a build needs a third hop / radix $\neq 12$ |
| P2 | OAM-trit saturates Holevo 1.585 bit/photon | a 3-mode channel misses/beats it |
| P3 | $[[66, 8, 3]]_3$ pseudo-threshold $\sim 5 \times 10^{-4}$ | encoding fails to help below it |
| P4 | magic robustness 3 / mana $\ln(5/3)$ | a lower-1-norm stabilizer decomposition |
| P5 | 5-Byzantine / 11-crash tolerance | 6 Byzantine survived, or 5 not |
| P6 | contextual fraction 1/10 ($\chi = 10 > 7$) | the benchtop witness $\hat{\chi} \leq 7$ |

Theorems (not falsifiable, only verifiable): $|\mathrm{Sp}(4, 3)| = 51840 = |W(E_6)|$; diameter 2, connectivity 12, bisection 100; distance 3; Clifford = $\mathsf{P}$ / +cubic = $\mathsf{BQP}$; 1-factorability; $\alpha = 7$, CF = 1/10.

**Honest scope**

The group orders, the bisection, the contextual fraction, $\alpha = 7$, the robustness $R = 3$, and the threshold coefficient $A$ are computed or measured in the public witnesses. The quantum values ($\mathsf{BQP}$, $\chi = 10$, Holevo saturation) assume the Witting orthonormal realisation in $\mathbb{C}^4$ and the standard quantum-computing theorems (Gottesman–Knill, Lloyd–Braunstein, Howard et al., Pashayan–Wallman–Bartlett). The $[[5, 1, 3]]_3$ code is a runnable stand-in for $[[66, 8, 3]]_3$ (same mechanism, different size). The QEC, teleportation, and threshold are exact / Monte-Carlo simulations (single-round; a full circuit-level treatment is future work). No physical photonic build exists yet; the quantum advantage is the priced $9^t$ dial, run here only for small $t$. "Architecture of life" means the von Neumann architecture of self-reproduction (the rigorous computational notion), not biology.

## 9 Conclusion

The Holonet is a complete finite computer architecture in which one 40-point geometry, one group, and one integer $q = 3$ generate the processor, the network, and the memory together; every layer is a theorem with a runnable witness, the device specification is its own audit, and the whole classical machine runs on any computer today. Its companion papers develop the physics (`photonic_holonet.tex`) and the infrastructural implications (`holonet_practical_implications.tex`); its quickstart is `HOLONET.md`. The first physical milestone — the benchtop contextuality test certifying the 1/10 fraction — is decided by a few hundred photons.

---

**Witnesses:** `analysis/w33_isa_encoding.py`, `w33_noc_floorplan.py`, `w33_one_group_machine.py`, `w33_complexity_advantage.py`, `w33_contextual_fraction.py`, `w33_ks_inequality.py`, `w33_magic_dial.py`, `holonet_node.py`, `holonet_qec_demo.py`, `holonet_teleport_demo.py`, `holonet_quine.py`, `holonet_quantum_packet.py`, `holonet_consensus_demo.py`, `holonet_threshold_demo.py`, `holonet_ks_experiment.py`, `w33_architecture_capstone.py`. **Standard results:** Gottesman–Knill; Lloyd–Braunstein; Howard–Wallman–Veitch–Emerson (2014); Pashayan–Wallman–Bartlett (2015); Cabello–Severini–Winter; Dolev; Lamport–Melliar-Smith; Thompson (VLSI); Bennett; Landauer.